

1 **Part 6: Heteroskedasticity**

2 Text references: §13.2, Chapter 14 in Ruppert and Matteson

3 Recall that for all stationary processes $V(X_t)$ is constant in t .

4 The models that we have studied to this point have further assumed
5 that the conditional variance $V(X_t|\mathcal{F}_{t-1})$ of the series is constant.

6 This assumption is called *conditional homoskedasticity*.

7 There are clearly many situations in which this is not a realistic as-
8 sumption, i.e. cases in which there is *heteroskedasticity*.

9 Here we will consider two approaches to dealing with this. The first
10 is simple *transformation* of the values of the series prior to modelling
11 the series. The second is the construction of models which directly
12 incorporate time-varying volatility. Examples of this second type in-
13 clude GARCH models and stochastic volatility models.

1 From Tsay (2010), page 111:

2 Volatility is not observable directly, but some characteristics
3 are commonly seen in stock price series:

- 4 • volatility clustering
- 5 • volatility evolves continuously with time, usually not with
6 jumps
- 7 • volatility does not diverge to infinity, in fact is approxi-
8 mately stationary
- 9 • volatility increases more following big price drops than it
10 does after big price increases – the *leverage effect*

Box-Cox Transformation

We first consider the *Box-Cox transformation* of the series.

Recall that we previously defined this transformation as $X^{(\lambda)}$ where

$$x^{(\lambda)} = \begin{cases} \frac{x^\lambda - 1}{\lambda}, & \text{if } \lambda \neq 0 \\ \log(x), & \text{if } \lambda = 0 \end{cases} \quad \lim_{\lambda \rightarrow 0} \frac{x^\lambda - 1}{\lambda} = \log(x)$$

This transformation is useful in cases where the variance of the process seems to increase with the value of the series. Consider the S&P 500 index during the period 1960 to 2017, shown in Figure 1.

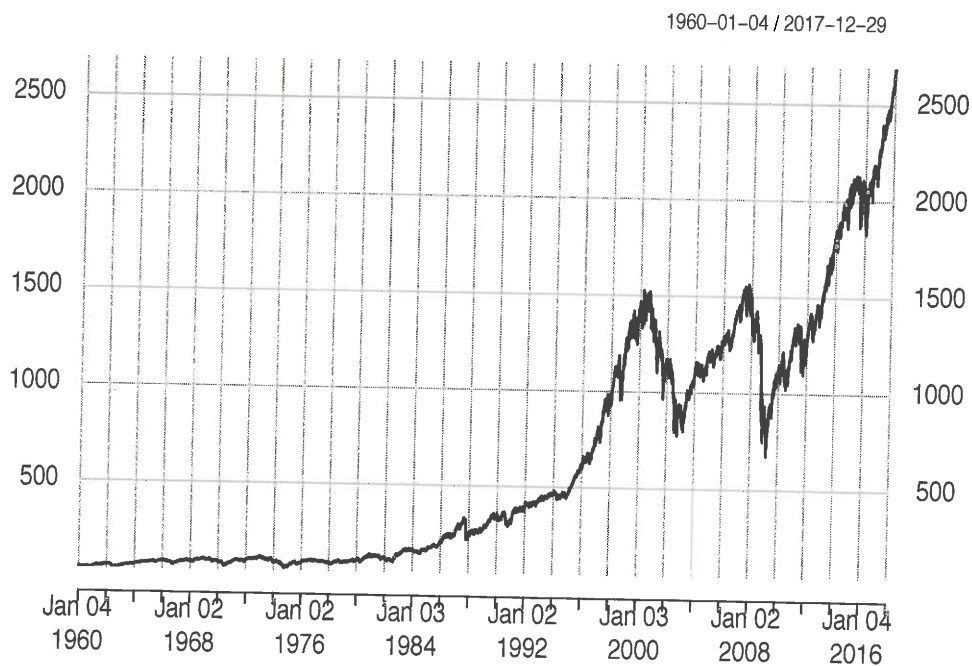


Figure 1: S&P 500 Index from 1960 through 2017.

Figure 2 shows the result of first differencing to this series. It should be clear that while the trend has been removed, there remains significant non-stationarity due to heteroskedasticity.

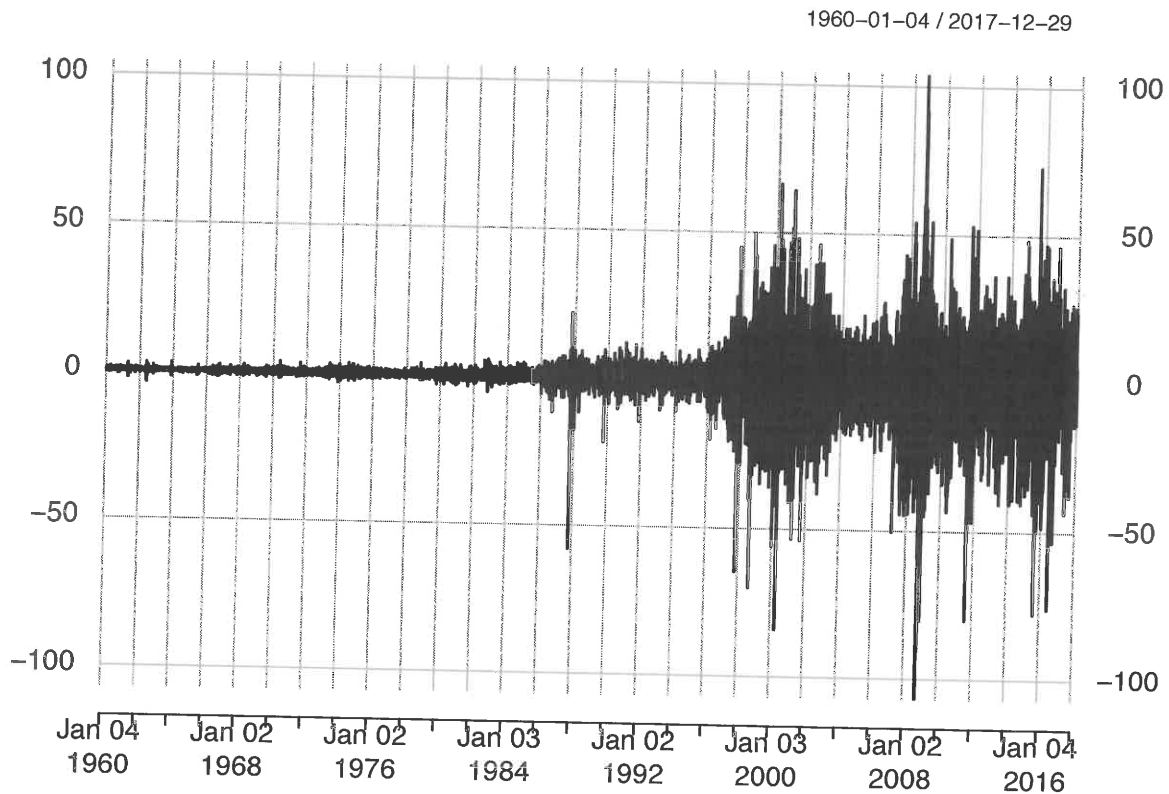


Figure 2: S&P 500 Index from 1960 through 2017, differenced.

In addition, it seems that the increased volatility is directly related to the increasing value of the series.

- 1 This would be a natural situation in which one would consider using
- 2 a transformation of the series. For example, one might take a log
- 3 transform (Box-Cox with $\lambda = 0$).
- 4 The result of this is shown as Figure 3. The improvement is evident.

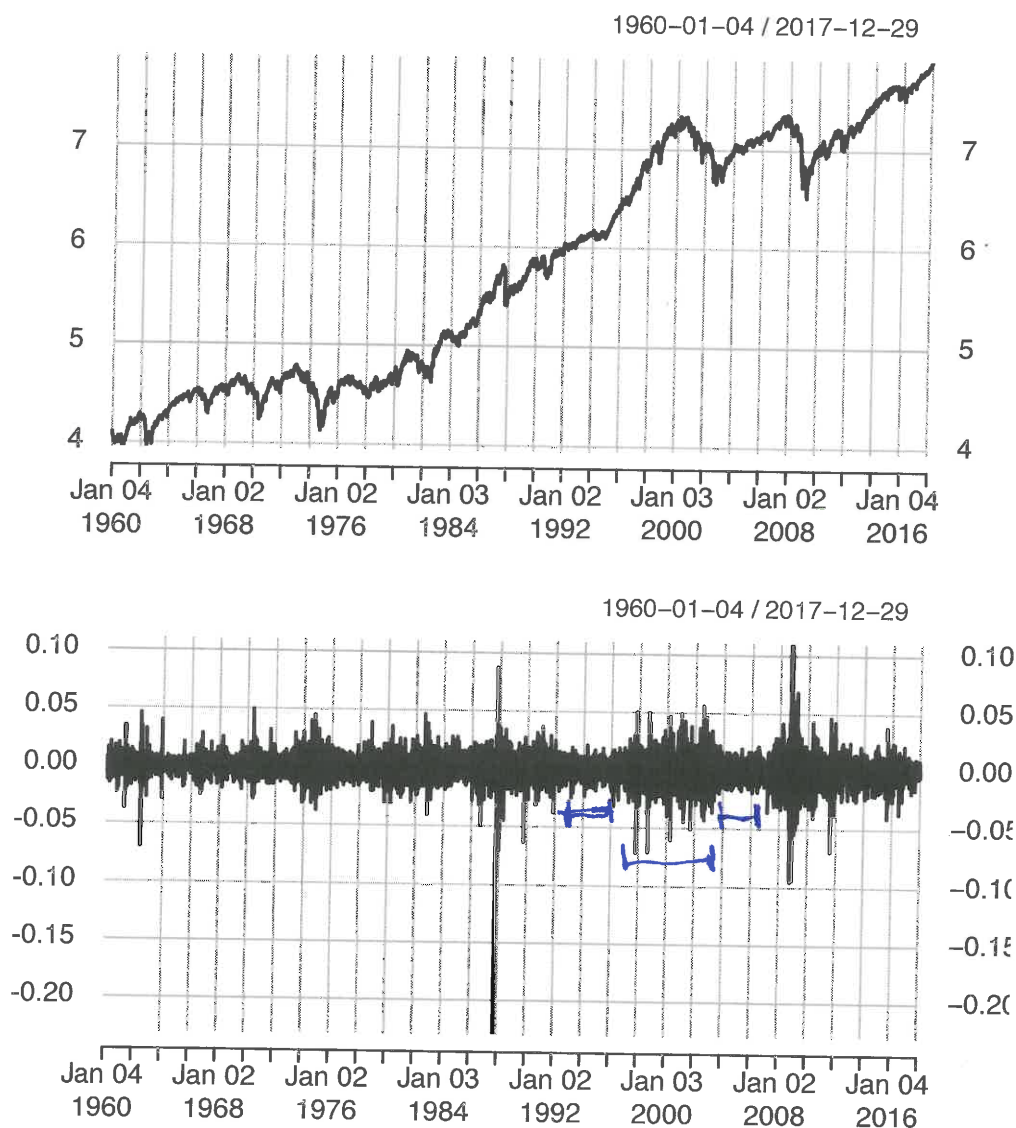


Figure 3: S&P 500 index from 1960 through 2017, log transformed, and then differenced.

1 Choosing the Value of λ

2 The value of λ in the Box-Cox transformation can be chosen in much
3 the same way it was chosen in regression.

4 The R package `FitAR` has a function `BoxCox.ts()` which will re-
5 turn the optimal transformation via a figure of the type of Figure 4.

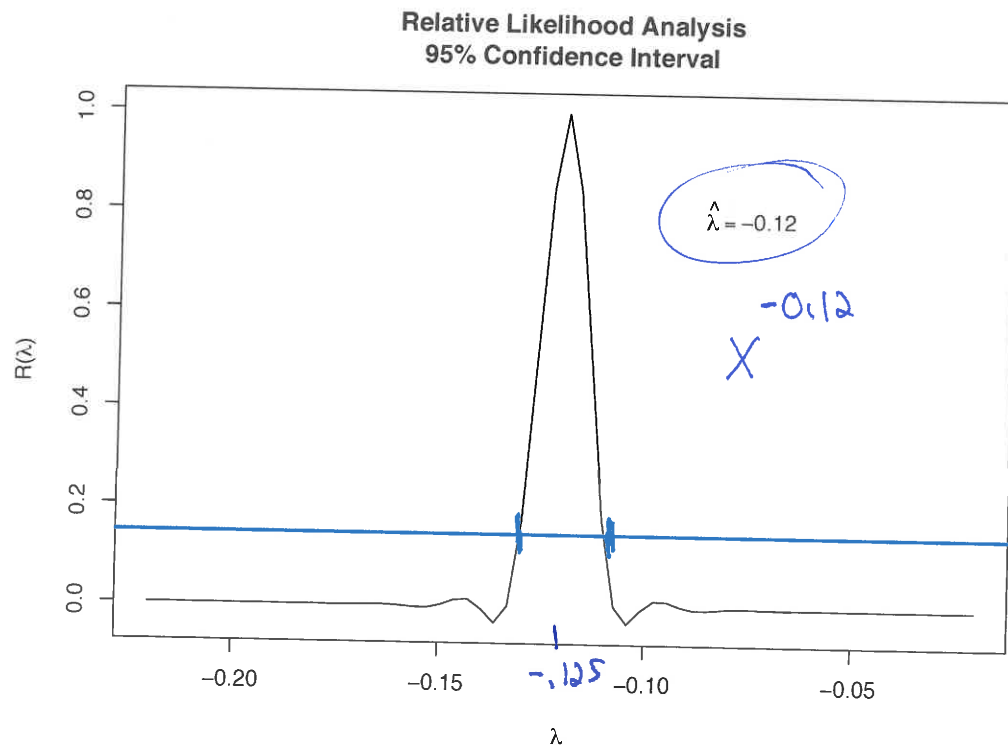


Figure 4: Relative likelihood for different values of λ , illustrating the choice of Box-Cox Transformation for the S&P 500 data.

1 The following commands will produce Figure 4.

```
2 > SP500 = getSymbols("SP500", src="FRED",  
3     auto.assign=F)  
4 > SP500 = SP500["1960::2017"]  
5 > SP500 = SP500[!is.na(SP500)]  
6  
7 > library(FitAR)  
8 > BoxCox.ts(as.numeric(SP500))
```

9 Note that SP500 must be wrapped with `as.numeric()`.

10 The transformed version of the series can be obtained by using the
11 `bxcx()` function (also part of package `FitAR`):

```
12 > SP500trans = bxcx(SP500, lambda = -0.12)
```

13 Then, models can be fit to this transformed version of the series.

14 Any forecasts can be transformed back to the original scale by us-
15 ing `bxcx()` with `InverseQ = TRUE`.

1

2 Overview of Conditional Heteroskedasticity

3 Although Figure 3 shows strong evidence that the variance of the
4 process has been approximately *stabilized*, there is still *volatility clus-*
5 *tering* evident: There are periods when the variance is smaller, and
6 then periods of time during which the variance is larger.

7 The Box-Cox transformation cannot help with this since it does not
8 incorporate any time-dependent transformation: The transformation
9 is only a function of the observed value of the series.

10 In particular, these models yield processes that are *conditionally het-*
11 *eroskedastic* (CH).

12 Examples of models which are conditionally heteroskedastic are the
13 GARCH models and stochastic volatility models.

1 For time series $\{X_t\}$, define

2
$$\mu_t = E(X_t | \mathcal{F}_{t-1})$$

3
$$\sigma_t^2 = V(X_t | \mathcal{F}_{t-1})$$

4 where $\mathcal{F}_{t-1}$ indicates the information that is available at time $t - 1$.

5 Recall that in the models considered to this point (e.g., ARMA), μ_t is
6 allowed to vary as a function of t , but σ_t^2 does not vary. For example,
7 for an AR(1) process,

8
$$\mu_t = \mu + \phi(X_{t-1} - \mu) = E(X_t | \mathcal{F}_{t-1})$$

9
$$\sigma_t^2 = \sigma_\epsilon^2 = V(X_t | \mathcal{F}_{t-1})$$

10 In conditionally heteroskedastic models, the opposite is true: The
11 value of μ_t is constant (usually zero) and σ_t^2 is allowed to vary.

1 How Does σ_t^2 Vary in These Models?

2 There are two major classes of models, depending on how σ_t^2 is al-
3 lowed to vary.

4 **Deterministic Models:** In these models the evolution of the value of
5 σ_t^2 is controlled by a deterministic equation: Given $\mathcal{F}_{t-1}$, one knows
6 the value of σ_t^2 exactly. GARCH models fall into this category.

7 **Stochastic Models:** In these models the evolution of the value of σ_t^2
8 has a random component, even when conditioned on $\mathcal{F}_{t-1}$. Stochas-
9 tic volatility models are in this category.

10 Note that “knowing σ_t^2 ” is not equivalent to knowing the value of
11 the series at time t . Our models will assume that the observed value
12 is $a_t = \sigma_t \epsilon_t$, where typically $\{\epsilon_t\}$ are iid from some distribution with
13 mean zero and unit variance.

1 How are These Models Used?

2 Models with conditional heteroskedasticity (CH) will get used in two
3 major ways:

4 **By Themselves:** In some situations, a time series model with con-
5 stant μ_t is a realistic model, and all that is needed is to model the
6 volatility appropriately. For example, the log returns of a stock are
7 often well-modelled by such a process with $\mu_t = 0$.

8 **As Part of a Larger Model:** In other situations, the CH process will
9 take the place of the white noise process in an ARIMA (or similar)
10 model. For example, one may assume that

$$11 \quad X_t = \mu + \phi(X_{t-1} - \mu) + a_t$$

12 where a_t is a ARCH(1) process (a particular CH process). Hence, one
13 can model both changes in μ_t and σ_t^2 .

GARCH Models

GARCH stands for “Generalized autoregressive conditionally heteroskedastic.”

These models will incorporate familiar elements of ARMA models, but applied to the squared values of the series instead of the series itself.

Here, we are going to let a_t denote the value of the series at time t , instead of the X_t we had been using to this point. The reason for this is that, as already mentioned, the GARCH model will often serve as a component in full model for the observed process $\{X_t\}$.

1 The ARCH(1) Process

2 Our starting point is the simple ARCH(1) model.

3 Assume that $\{\epsilon_t\}$ is iid white noise with mean zero and unit variance.

4 Define

5
$$a_t = \epsilon_t \sqrt{\omega + \alpha_1 a_{t-1}^2}$$

6 where $\omega > 0$ and $\alpha_1 \geq 0$. Then $\{a_t\}$ is an ARCH(1) process.

7 We have already shown (Part 1) that for the ARCH(1) model,

8
$$\sigma_t^2 = \omega + \alpha_1 a_{t-1}^2$$

9 and that if $\alpha_1 < 1$, then $\{a_t\}$ is stationary.

10 We also showed that, in the case $\alpha_1 < 1$, the series $\{a_t\}$ is white noise
11 with mean zero and variance $\omega/(1 - \alpha_1)$.

- 1 Figure 5 compares simulated series of length $n = 200$ generated from
- 2 different ARCH(1) processes.

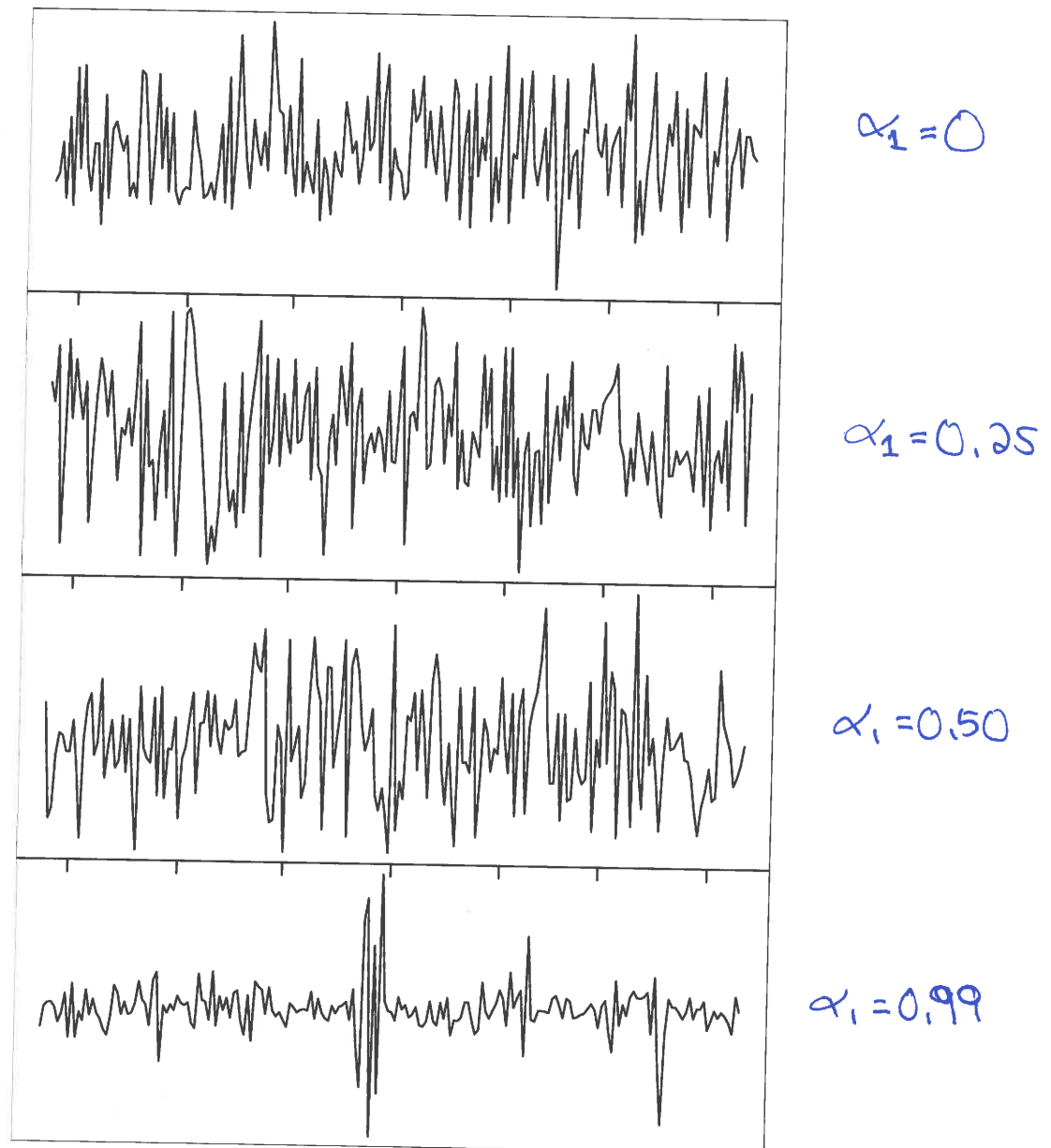


Figure 5: Simulated ARCH(1) processes. In each case $n = 200$ and $\omega = 0.01$. From top to bottom, $\alpha_1 = 0, 0.25, 0.5, 0.99$.

1 The ARCH(p) Process

2 The ARCH(1) model can be extended to the ARCH(p) model by writ-
3 ing that

$$4 \quad a_t = \epsilon_t \sqrt{\omega + \sum_{i=1}^p \alpha_i a_{t-i}^2}$$

5 where $\{\epsilon_t\}$ is iid white noise with unit variance. It is assumed that
6 $\omega > 0$ and $\alpha_i \geq 0$ for all i .

7 Note that this implies that

$$8 \quad \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2.$$

9 The ARCH(p) process also generates a stationary, uncorrelated se-
10 quence $\{a_t\}$ under appropriate conditions on the α_i .

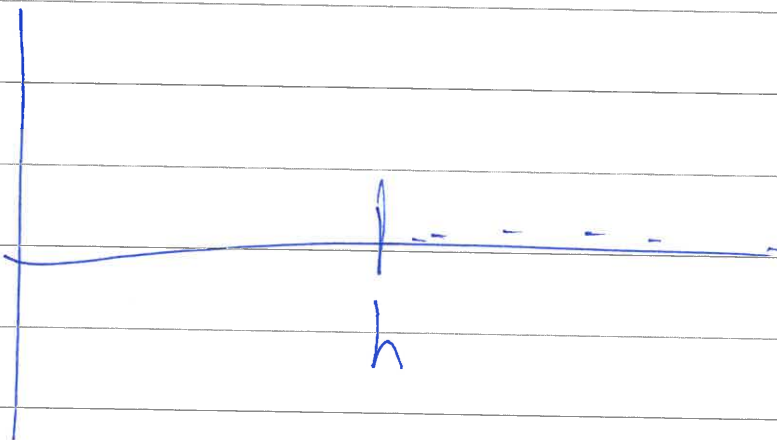
11 It is also possible to show that the sequence $\{a_t^2\}$ has the same ACF
12 and PACF of an AR(p) process.

13 **Exercise:** Describe the practical importance of this result.

14 ① Use Ljung-Box Test on $\{a_t^2\}$ to
15 judge if ARCH component needed

② Use the PACF of $\{a_t^2\}$ to help choose the value of p

If PACF close to 0 for lags greater than h , choose $p \leq h$



1 Fitting an ARCH(p) Model

2 The GARCH models we will discuss can be fit using the function

3 `garchFit()` which is part of the package `fGarch`.

4 Consider the following example of the fit of an ARCH(2) model to a

5 sequence of log return data:

```
6 > MSFT = getSymbols("MSFT", from="2010-1-1",  
7                       to="2017-12-31", auto.assign=F)  
8 > logret = dailyReturn(MSFT, type="log")  
9 > holdfit = garchFit(~garch(2,0), data=logret,  
10                      include.mean=FALSE)
```

11 Comments:

- 12 • The "2" in `garch(2,0)` is where one specifies the value of p .
- 13 • Note the use of the "tilde" prior to `garch(2,0)`.
- 14 • By setting `include.mean=FALSE`, we force the process to as-
- 15 sume a mean zero.

1 The output of summary(holdfit) includes a wealth of informa-
 2 tion, including the following:

3 Error Analysis:

	Estimate	Std. Error	t value	Pr(> t)
4 omega	0.0001628 $\hat{\omega}$	0.0000077	21.141	< 2e-16 ***
5 alpha1	0.1012654 $\hat{\alpha}_1$	0.0318502	3.179	0.00148 **
6 alpha2	0.0902323 $\hat{\alpha}_2$	0.0334075	2.701	0.00691 **

$H_0: \theta = 0$
 vs. $H_1: \theta \neq 0$

8 ---

9 Signif. codes: 0 *** 0.001 ** 0.01 * 0.05 . 0.1 1

10

11 Log Likelihood:

12 5751.038 normalized: 2.856949

13

14 Information Criterion Statistics:

	AIC	BIC	SIC	HQIC
15	-5.710917	-5.702561	-5.710922	-5.707850

1 Residuals

2 The default of `garchFit()` is to model the ϵ_t as being standard nor-
3 mal. We will describe later how this can be adjusted.

4 As before, we can assess the validity of the normality assumption by
5 looking at the normal probability plot of the residuals, but one needs
6 to be careful with the definition of the residuals in this case.

7 Let $\{x_t\}$ denote the original series. The *raw, unstandardized residuals*
8 are defined as

9
$$\hat{e}_t = \underset{\text{obs}_t}{x_t} - \hat{\mu}_t \text{ for all } t.$$

10 These are the residuals returned by

11 `> residuals(holdfit)`

- 1 **But**, the raw residuals are not approximations to the ϵ_t .
- 2 Indeed, if a zero mean is assumed for the process, the raw residuals
- 3 are identical to the series itself.

4 Instead, we define the *standardized residuals* as

$$5 \quad \hat{\epsilon}_t = \hat{e}_t / \hat{\sigma}_t = (x_t - \hat{\mu}_t) / \hat{\sigma}_t$$

6 **Exercise:** Justify this definition for the residuals within the context of

7 the ARCH(p) model.

8 If we fit ARCH(p) model with $\mu_t = 0$,

9 then we are assuming

$$10 \quad a_t = \epsilon_t \sqrt{\omega + \sum_{i=1}^p \alpha_i a_{t-i}^2} = \epsilon_t \sigma_t$$

$$12 \quad \text{So, } \epsilon_t = a_t / \sigma_t$$

15 seems reasonable to define $\hat{\epsilon}_t = a_t / \hat{\sigma}_t$

- 1 The estimates of σ_t can be found in R from
- 2 `> holdfit@sigma.t`
- 3 so the standardized residuals are
- 4 `> sresids = residuals(holdfit)/holdfit@sigma.t`
- 5 **R Exercise:** Continue the example that we have begun. Verify that
- 6 $p = 2$ was a good choice for this model. Find the vector of standard-
- 7 ized residuals. This model has been fit under the assumption that
- 8 the ϵ_t are normal; is that assumption justified?

The PACF shows evidence of correlation at lag 1 (and maybe even at lag 25!). The choice $p=2$ may have been too much.

The normal prob. plot of standardized residuals shows significantly heavier tails than normal would predict

1 The GARCH(p, q) Model

2 In our simulated ARCH(1) processes (Figure 5), we saw that the vari-
3 ability and degree of persistence in the volatility was not that great.
4 GARCH(p, q) models will typically have more flexibility to fit this
5 type of behavior.

6 Again, let $\{\epsilon_t\}$ be iid white noise with unit variance. Then, we can
7 write

$$8 \quad a_t = \sigma_t \epsilon_t$$

9 where

$$10 \quad \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2.$$

11 Here, it is assumed that $\omega > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, and

$$12 \quad \sum_{i=1}^{\max\{p,q\}} (\alpha_i + \beta_i) < 1.$$

13 These conditions guarantee that each a_t has finite variance, and sta-
14 tionarity.

- 1 **Exercise:** What feature of this model going to allow for more persis-
2 tence in the volatility?

3 In ARCH, σ_t^2 depends only on previous
4 values in series. If $p=2$, and by
5 chance see two values in a row close
6 to 0, even if σ_t^2 are relatively large
7 with GARCH, σ_t^2 depends on previous
8 values of $\{\sigma_t^2\}$ as well. These
9 values are more stable.

- 10
11
12 Note that the ARCH(p) model is the GARCH(p, q) model with $q = 0$.

- 13 **Exercise:** What would happen if $p = 0$?

14 There is no random fluctuations when
15 $p=0$. Need $p \geq 1$.

1 Consider the behavior of the GARCH(p, q) via simulations.

2 The R package `fGarch()` includes the function `garchSim()`. Con-
3 sider the following example:

```
4 > simout = garchSim(n=200,  
5   garchSpec(list(omega=0.1, alpha=0.3, beta=0.5)),  
6   extended=T)
```

7 In this example:

- 8 • The length of the series is $n = 200$.
- 9 • This is a GARCH(1,1) model with parameters set to $\omega = 0.1$, $\alpha_1 =$
10 0.3 , and $\beta_1 = 0.5$.
- 11 • By setting `extended=T`, the function returns not only the simu-
12 lated series, but also the sequence of values of σ_t and the simu-
13 lated ϵ_t values.

14 An example of the output:

```
15           garch      sigma      eps  
16 2018-03-05 -0.48988710 0.4706097 -1.04096263  
17 2018-03-06  0.62021376 0.5317270  1.16641398  
18 2018-03-07 -0.04370719 0.5972992 -0.07317470  
19 ...
```

1 Figure 6 shows the result of

2 `> plot(simout)`

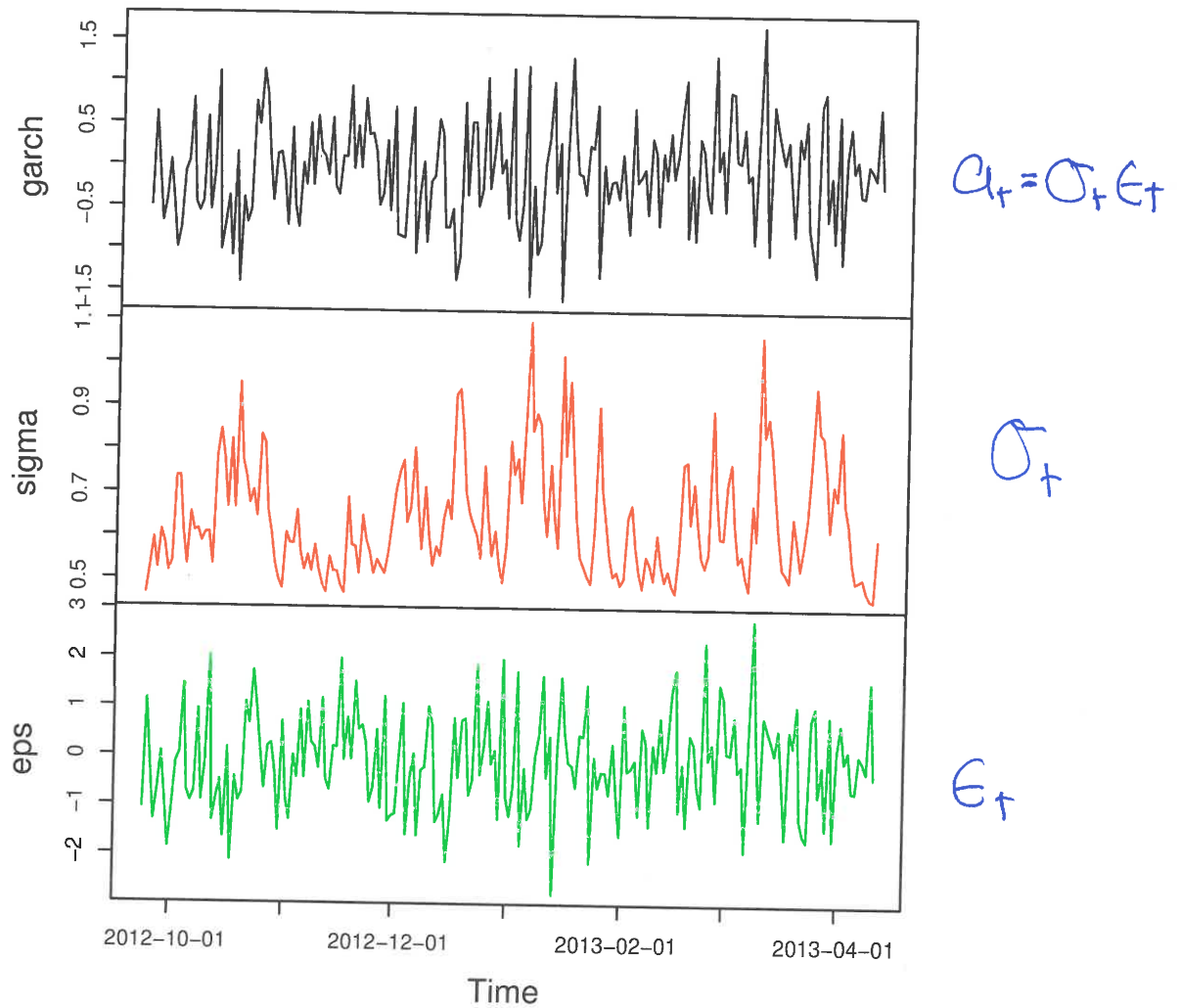


Figure 6: Output of `garchSim()`.

ARMA Representation of GARCH(p,q)

The GARCH(p,q) model is such that

$$a_t = \sigma_t \epsilon_t$$

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$$

Define $n_t = a_t^2 - \sigma_t^2$ $\sigma_t^2 = a_t^2 - n_t$

$$\begin{aligned} E(n_t | \mathcal{F}_{t-1}) &= E(a_t^2 - \sigma_t^2 | \mathcal{F}_{t-1}) \\ &= E(a_t^2 | \mathcal{F}_{t-1}) - E(\sigma_t^2 | \mathcal{F}_{t-1}) \\ &= \sigma_t^2 E(\epsilon_t^2 | \mathcal{F}_{t-1}) - \sigma_t^2 \\ &= \sigma_t^2 V(\epsilon_t) - \sigma_t^2 \\ &= 0 \end{aligned}$$

So, $\{n_t\}$ is a mean zero MDS

Now, rewrite GARCH model as

$$a_t^2 = \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + n_t - \sum_{j=1}^q \beta_j n_{t-j}$$

So, a_t^2 modelled as a function of
previous a_{t-i}^2 values (AR component)
and the $\{n_t\}$ series (MA component)

This is an ARMA($\max(p, q)$, q) model
 $\{a_t^2\}$. The $\{n_t\}$ are the "innovations"

1 Fitting a GARCH(p, q) Model

2 According to Tsay: “[S]pecifying the order of a GARCH model is not
3 easy. Only lower order GARCH models are used in most applica-
4 tions, say, GARCH(1,1), GARCH(1,2), and GARCH(2,1) models.”

5 The fitting of GARCH(p, q) model is done in R using `garchFit()`,
6 but now with `garch(p, q)` specified.

7 For example, if we wanted to fit a GARCH(1,1) model to our earlier
8 example, the command would be

```
9 > holdfit = garchFit(~garch(1,1), data=logret,  
10     include.mean=FALSE)
```

1 The Distribution of $\{\epsilon_t\}$

2 To this point, we have assumed that the ϵ_t are iid with mean zero and
3 unit variance; the default in `garchSim()` and `garchFit()` is to
4 assume that they are normal. This can be adjusted in both functions
5 using the argument `cond.dist`.

6 There are three choices for distributions symmetric about zero:

- 7 • **Standard Normal** (`cond.dist="norm"`)
- 8 • **Standard t** (`cond.dist="std"`): This is the distribution which
9 we have considered previously, but scaled to have a variance of
10 one. There are functions `dstd()`, `pstd()`, `qstd()`, and `rstd()`;
11 each has argument `nu` which controls the number of degrees of
12 freedom. Note that since the variance must equal one, the de-
13 grees of freedom must be at least three.

$$\frac{\nu}{\nu-2}$$

- **Generalized Error Distribution (GED)** (`cond.dist="ged"`):

This is another family of distributions symmetric about zero, and with unit variance. It can create pdfs with more “pointed” peaks, and also includes the standard normal and the double exponential distributions as special cases. The shape is controlled by the parameter ν , often called the “shape parameter.” There are functions `dged()`, `rged()`, and so forth, each of which includes `nu` as an argument.

Figure 7 shows the pdf of the GED as a function of ν . See Section 5.6 in Ruppert and Matteson for more details.

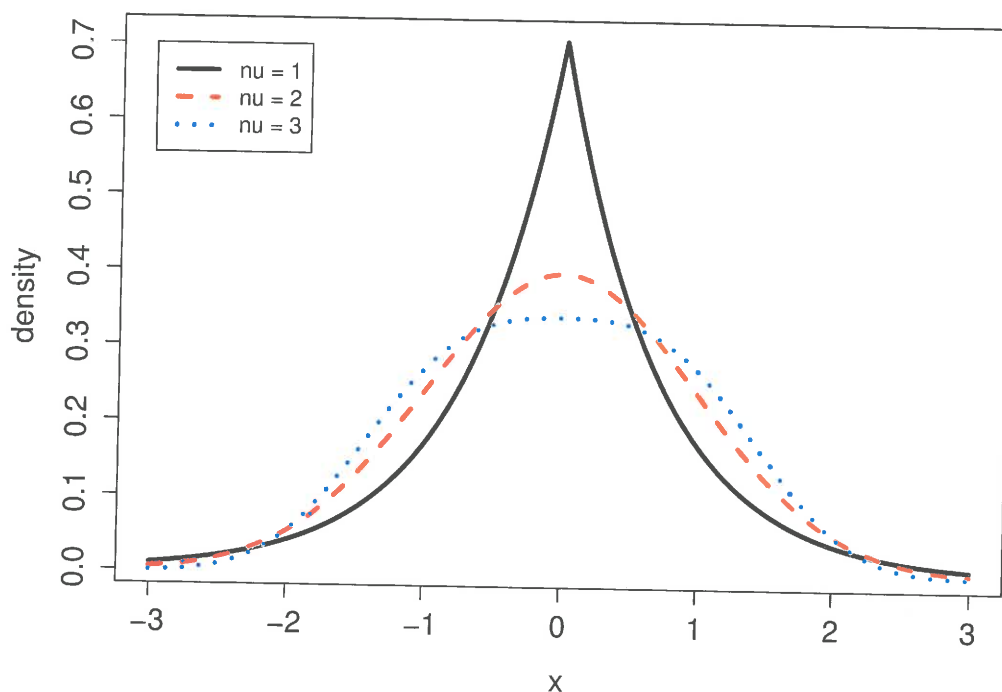


Figure 7: The GED, for different values of ν .

1 Each of the above three distributions has an *skewed* version that can
2 be utilized.

3 As described in Section 5.7 of Ruppert, if one starts with a pdf f
4 which is symmetric about zero, then a skewed version can be con-
5 structed by initially constructing

$$6 \quad f^*(x|\xi) = \begin{cases} f(x\xi), & \text{if } x < 0 \\ f(x/\xi), & \text{if } x \geq 0 \end{cases}$$

7 Then, $f^*(x|\xi)$ is scaled and shifted so that it has mean zero and unit
8 variance; the result is the skewed version of the density f .

9 Note that there is a new parameter ξ which controls the amount of
10 skew. It must hold that $\xi > 0$. The symmetric case is $\xi = 1$.

11 Also note that ξ and $1/\xi$ yield an equivalent amount of skew; one
12 case is the mirror image of the other, reflected about zero.

- 1 Figure 8 shows the skewed standard normal distribution for different
2 values of ξ . Keep in mind that each of these distributions has mean
3 zero and unit variance.

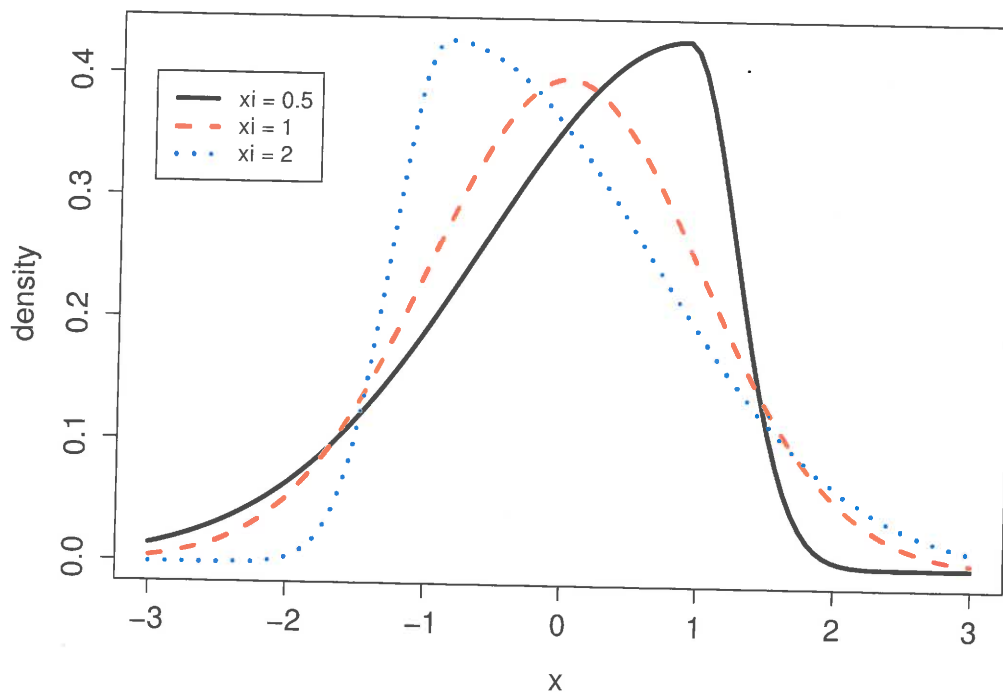


Figure 8: The skewed standard normal, for different values of ξ .

- 4 In R, package `fGarch()` includes functions `dsnrm()`, `dsstd()`,
5 `dsGED()`, `rsnorm()`, `rsstd()`, `rsGED()`, and so forth.
- 6 And when using `garchFit()` or `garchSim()`, set `cond.dist` to
7 `snorm`, `sstd`, or `sGED`.

1 We have seen how the normal probability plot can be used to judge
2 the validity of the normality assumption; this idea can be extended to
3 testing any distribution, provided it is possible to simulate a sample
4 from that distribution.

5 The generalization is called a *quantile-quantile plot* or *QQ plot*.

6 As with the normal probability plot, the goal is to plot the quantiles
7 of the observed data versus the quantiles under the distribution be-
8 ing considered.

1 The R function `ppoints()` creates a list of numbers evenly spaced
2 between 0 and 1; the only argument required is the length of the
3 series. For example:

```
4 > ppoints(5)
5 [1] 0.1190476 0.3095238 0.5000000 0.6904762
6 [5] 0.8809524
```

7 Now, use one of the `q*` functions to get the vector of quantiles for
8 the distribution under consideration:

```
9 > qstd(ppoints(5), nu=7)
10 [1] -1.0900787 -0.4395454 0.0000000 0.4395454
11 [5] 1.0900787
```

12 In this example, I'm using the standard t distribution with seven de-
13 grees of freedom.

14 **Exercise:** Suppose my data set consists of n values. What should be
15 expected if the sorted values of data versus a series constructed in the
16 above manner, and the theoretical distribution is chosen correctly?

17 Should see (roughly) a straight line
18
19
20

- Figure 9 illustrates the idea. Here, a sample of size $n = 1000$ was generated from the standard t distribution with four degrees of freedom. The figure shows a sequence of four QQ plots, each with a different theoretical distribution for comparison.

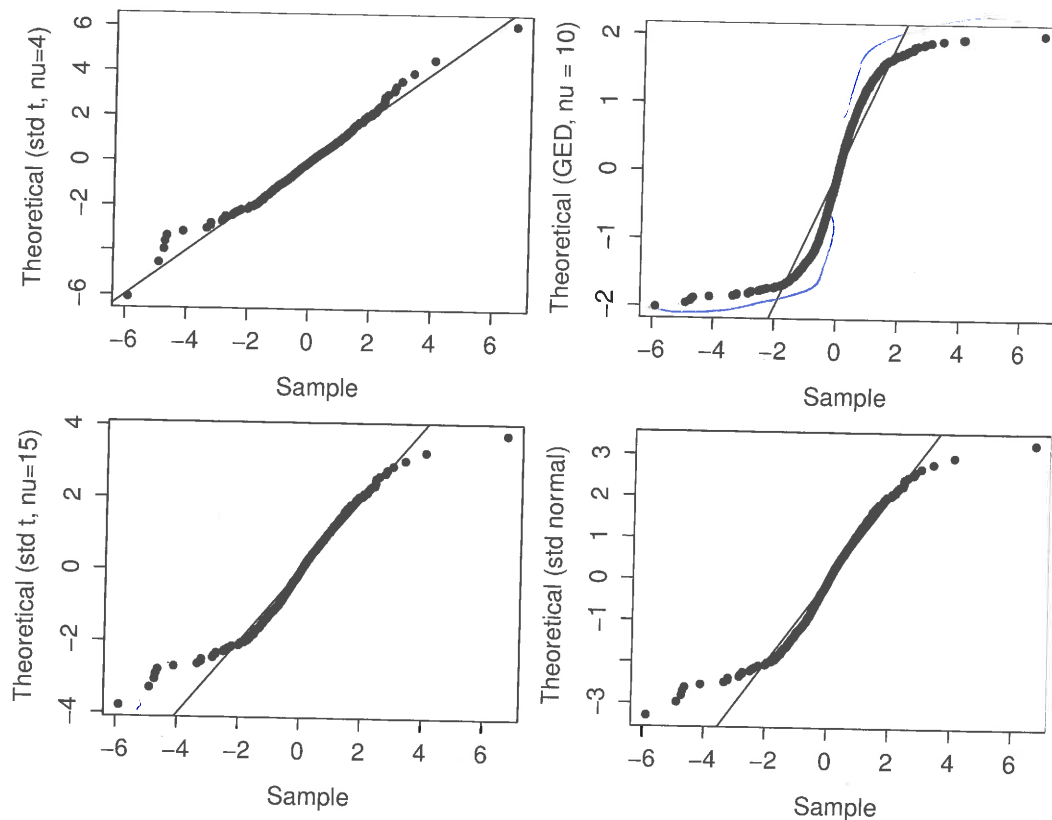


Figure 9: QQ plots comparing a simulated sample.

Exercise: What is the conclusion from Figure 9?

Seems that t w/ 4 d.f. is a decent fit

R Exercise: Obtain the daily log returns for SPY for 2010 through 2017. Compare the fit of GARCH(1,1) models under the assumption that ϵ_t are (1) standard normal, (2) standard t, and (3) skewed standard t.

The initial commands are shown below.

```

6 > SPY = getSymbols("SPY", from="2010-1-1", to="2017-12-31",
7     auto.assign=F)
8 > logret = dailyReturn(Ad(SPY), type="log")
9 > holdfit1 = garchFit(~garch(1,1), dat=logret, include.mean=F)
10 > holdfit2 = garchFit(~garch(1,1), dat=logret, include.mean=F,
11     cond.dist="std")
12 > holdfit3 = garchFit(~garch(1,1), dat=logret, include.mean=F,
13     cond.dist="sstd")

```

The normal assumption seems unwarranted

The standard t dist. definitely improves the fit.

The skewed t dist. improves the fit on the upper tail.

$\hat{\xi} = 0.87$ with SE of 0.023, which suggests strong evidence of skew

AIC values: For symmetric: -6.884 supports
 For skewed: -6.89 including skew

The APARCH(p, q) Model

In the modelling of time series of financial returns, an often-apparent feature is that large negative returns increase volatility more than do large positive returns. This is called the *leverage effect*.

The standard GARCH model does not allow for the leverage effect.

Recall that this model assumed that

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2.$$

Hence, the value of σ_t^2 is a function of the square of the previous values of the series. Of course, the square is a symmetric function.

APARCH stands for “asymmetric power ARCH.” With this class of models, a skewed function of different powers of σ_t will be modelled.

New parameters are introduced that control the amount of skew and the power that is utilized.

The APARCH(p, q) model can be written as follows:

$$\sigma_t^\delta = \omega + \sum_{i=1}^p \alpha_i (g_{\gamma_i}(a_{t-i}))^\delta + \sum_{j=1}^q \beta_j \sigma_{t-j}^\delta$$

where

$$g_\gamma(x) = |x| - \gamma x,$$

and it holds that $\delta > 0$ and $-1 < \gamma_i < 1$.

Figure 10 shows how $g_\gamma(x)$ varies as a function of γ .

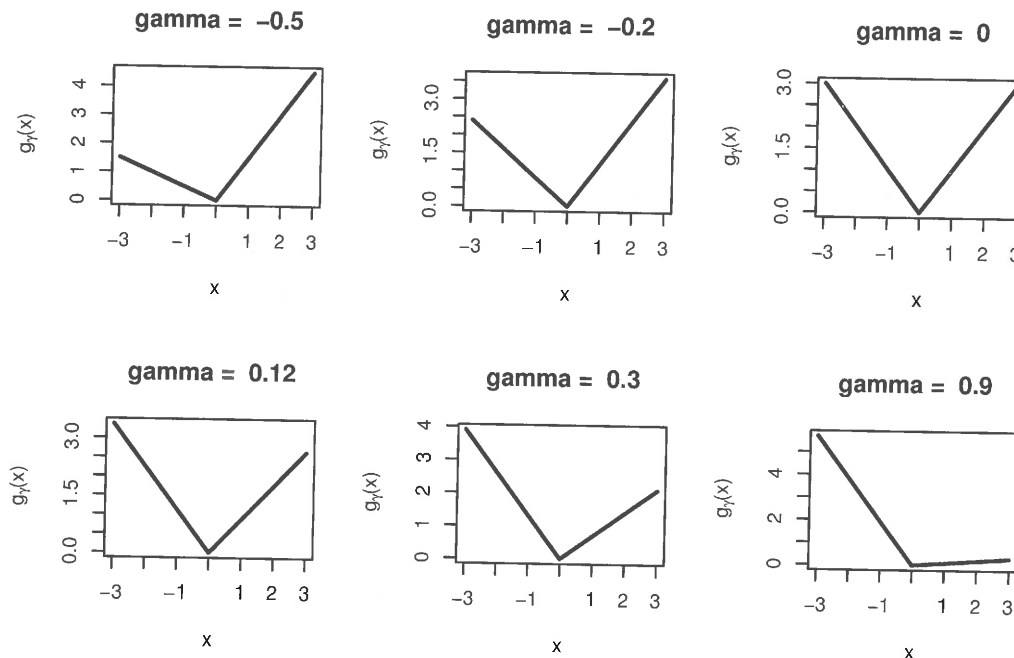


Figure 10: Reproduced from Ruppert, Figure 14.8. Figure shows how $g_\gamma(x)$ varies as a function of γ .

1 Some comments on this model:

- 2 • There is a separate skew parameter γ_i for each of the lags $i =$
3 $1, 2, \dots, p$.
- 4 • If $\delta = 2$ and $\gamma_i = 0$ for all i we get back the standard GARCH
5 model.
- 6 • When $\gamma > 0$, it follows that $g_\gamma(-x) > g_\gamma(x)$, hence indicating the
7 presence of a leverage effect.
- 8 • For the same time series, values of AIC can be compared between
9 GARCH and APARCH fits. This is a useful way to judge the need
10 for this additional complexity.
- 11 • In R, one fits an APARCH(p, q) model by using the argument
12 `aparch(p, q)` in the function `garchFit()`. This is used in
13 place of `garch(p, q)`.
- 14 • In some situations it is best to fix $\delta = 2$ (or maybe even some
15 other value). This can be accomplished by setting argument `delta=2`
16 along with `include.delta=FALSE`.

1 **R Exercise:** Fit an APARCH(1,1) model to the daily log returns on
2 Ford for the period 1995 to 2017. Is there strong evidence that the
3 complexity of the APARCH is necessary? Compare also with the
4 case where δ is fixed to be 2. Be sure to check the distribution of the
5 standardized residuals.

```
6 > FORD = getSymbols("F", from="1995-1-1",  
7                       to="2017-12-31", auto.assign=F)  
8 > logret = dailyReturn(Ad(FORD), type="log")
```

1 The EGARCH(p, q) Model

2 A close relative to the APARCH model is the *exponential GARCH*
3 (*EGARCH*) model:

$$4 \log(\sigma_t^2) = \omega + \sum_{i=1}^p \alpha_i (g_{\gamma_i}(a_{t-i})) / \sigma_{t-i} + \sum_{j=1}^q \beta_j \log(\sigma_{t-j}^2)$$

5 This is very similar to an APARCH model where δ is chosen to be
6 close to zero.

Combining CH and ARIMA Models

As already mentioned, one use for conditionally heteroskedastic models is to replace what previously was typically taken to be iid white noise in ARIMA models.

Consider the following example. Suppose that $\{\epsilon_t\}$ are iid with mean zero and unit variance. The AR(1) model can be written as

$$X_t = \mu + \phi(X_{t-1} - \mu) + \sigma_\epsilon \epsilon_t$$

To construct the AR(1)/ARCH(1) model we write

$$X_t = \mu + \phi(X_{t-1} - \mu) + \epsilon_t \sqrt{\omega + \alpha_1 a_{t-1}^2}$$

Exercise: We previously derived/presented properties of the AR(1) process. Do these need to be updated for the AR(1)/ARCH(1) model?

No. Those derivations were done under the assumption that ϵ_t were white noise. The ARCH models are also white noise.

1 The above generalizes: We can combine any of the ARIMA models
2 with a GARCH (or APARCH) model to create a model that has both
3 varying μ_t and varying σ_t .

4 In R, this is accomplished easily using the function `garchFit()`.

5 Include the $\text{ARMA}(p, q)$ term and $\text{GARCH}(p_g, q_g)$ terms together as

```
6 > holdfit = garchFit(~arma(p,q)+garch(pg,qg),  
7       data=logret)
```

8 Comments:

- 9 • It is important that `arma(p,q)` comes before `garch(pg,qg)`.
- 10 • It is not be typical to use `include.mean=F` when the ARMA
11 portion is included.
- 12 • `garch(pg,qg)` can be replaced with `aparch(pg,qg)`.
- 13 • If you want to include differencing, then this has be done manu-
14 ally to the series prior to calling this function.
- 15 • It is required that a GARCH or APARCH component be included;
16 to fit a model without such a component, use `arima()`.

- 1 **Exercise:** Suppose one is fitting an ARIMA model. What evidence
2 could be used to indicate the need for a GARCH component?

3 Look at the ACF of the squared
4 residuals from the ARIMA fit. If
5 there is vol. clustering it should
6 appear as serial correlation in the
7 squared residuals.

- 11 **Exercise:** Can ARMA models be compared with ARMA/GARCH
12 models using AIC?

13 Yes. As long as the series being
14 modelled stays the same.

1 **Comment on AIC Output by `garchFit()`:**

2 The value of information criteria can be accessed using

3 `> attributes(holdfit)$fit$ics`

4 Also, note that `garchFit()` standardizes these quantities by divid-
5 ing by the length of the series. To compare the AIC returned by this
6 function with that returned by `arima()`, one must multiply the AIC
7 returned by `garchFit()` by the length of the series.

The output of `garchFit()` includes a table like that below.

Standardised Residuals Tests:

			Statistic	p-Value	
Jarque-Bera Test	R	Chi ²	5195.223	0	H ₀ : ϵ_t drawn from normal
Shapiro-Wilk Test	R	W	0.9278109	0	
Ljung-Box Test	R	Q(10)	10.00205	0.4403137	Need more complex model for μ_t ?
Ljung-Box Test	R	Q(15)	14.98568	0.4524491	
Ljung-Box Test	R	Q(20)	22.00736	0.3401116	
Ljung-Box Test	R ²	Q(10)	3.399541	0.9703998	Need more complex model for σ_t^2 ?
Ljung-Box Test	R ²	Q(15)	5.063953	0.991571	
Ljung-Box Test	R ²	Q(20)	7.103106	0.996333	
LM Arch Test	R	TR ²	4.262976	0.9782099	

The first two are tests for normality of the ϵ_t . The next six are application of the Ljung-Box Test to the standardized residuals (R) and the standardized residuals squared (R²).

The final test is the *Lagrange Multiplier Test*.

Each of the last four tests are used to test for the need of additional ARCH complexity to the model. They are testing for remaining serial correlation in the squared standardized residuals.

- 1 **R Exercise:** Continue the Ford example started above. Compare
- 2 models that include an ARMA component with models which ex-
- 3 clude it. Compare APARCH models with GARCH models, also.